Mohammad Uzair Khan

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Education

BITS Pilani, Dubai Campus — Dubai, UAE
B.E. Computer Science — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

Profile

Computer Science undergraduate building **AI agents**, **automations** and **data systems** that do real work, including an agent I run on my own inbox every day: LLMs that **read, classify, extract and draft** with **structured outputs** and held-out evaluation; **data pipelines**, **SQL** and forecasting; dashboards in **Power BI** and React; integrations over **REST APIs, webhooks and scheduled jobs**. Every result is measured against a baseline and reported even when it disappoints.

AI Agents, Automation & Data Projects

Inbound: Scout, the inbox agent I built for myself | *Python, Flask, SQLite, Gmail API, React* — Live | Code
- Reads my university **Gmail** live (read-only), **classifies** every message, **extracts** internship calls and college deadlines, and **drafts** a CV pack from approved evidence; I review and submit
- One agent with a live step journal, **memory**, rules **learned from my corrections** and a cited chat; sorted **116/116** held-out emails and read 365 of 387 fields blind; **617** pytest, **92** Playwright tests

Backroom: AI back-office agent for a restaurant group | *Python, Gemini API, PostgreSQL, Flask, React* — Live | Code
- LLM agent reads invoice PDFs, emails and payout statements into **typed records** (JSON schema, source quotes, rule validation): **99.8%** of fields on unseen layouts vs **12.7%** for rules only; **0 of 221** accepted documents carry a wrong amount
- **Cleans names** (precision 1.0, recall 0.925), **reconciles** POS, payouts and ledger in **SQL** (31/31 planted problems, 0 false alarms), estimates **28.8 hours saved**; **tool-calling** Ask agent 29/30, 0 wrong

Agent Desk: AI agents for GitHub and LinkedIn | *Python, FastAPI, Claude, Next.js* — Live | Code
- Two **Claude** agents (via the **Claude Code** CLI) with **JSON-schema output**: rules first, at most one model call, nothing written without a human click; a **scheduled job** audits repos every six hours

Rx Market Pulse: data pipeline and dashboards | *Python, SQL (DuckDB), Power BI* — Live | Code
- Tested **pipeline** over **36.1M** public rows; **harmonized 1,488** name variants with **19** data-quality checks; a **Power BI** report (star schema, 28 DAX measures) and a web dashboard share one set of numbers

Brand Forecast Studio: sales forecasting | *Python, LightGBM, Pydantic, TypeScript* — Live | Code
- One global **LightGBM** model on **13,336** brand-quarters; rolling-origin backtest WAPE **17.9%** vs 19.4% for the best baseline; **conformal** intervals lift coverage from 70% to **79.6%**; PSI drift checks

Also built: Order Desk · LoopDesk + FollowDesk *(CRM)* · Invoice Ready · SignalDesk AI · GCC Market Report — GitHub

Experience

Backend Developer Intern, Amaani (Travel-Tech Startup) — Dubai, UAE | *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

Research & Publications

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices — 2025 – 2026
First Author; submitted to IEEE MSN 2026 (under review)
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

Technical Skills & Certification

AI & Agents: Gemini API; Claude via Claude Code; structured outputs (JSON Schema, Pydantic), tool calling, RAG (BM25), held-out evaluation
Data & ML: Python, SQL (PostgreSQL, SQLite, DuckDB), pandas, LightGBM, forecasting, data cleaning, pipelines, Power BI (DAX), Excel
Integration: Flask, FastAPI, REST APIs, webhooks, scheduled jobs (GitHub Actions), Gmail API, PDF/email parsing
Front end & tools: TypeScript, React, Next.js, pytest, Playwright, Git, CI/CD, Vercel, Neon; Claude Code and Codex
Certification: The Ultimate Job Ready Data Science Course, CodeWithHarry (2026)